ANGUS CHOU

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EDUCATION

University of British Columbia | Computer Science (Co-op) **Sept 2025 - Expected Jun 2028**
Coursework: Computer Graphics (CPSC 314), Object-Oriented Programming, DSA, OOP

TECHNICAL SKILLS

Languages: C++, C#, Java
Game Engines & Tools: Unreal Engine 5, Unity, Godot, Visual Studio, Git, GitHub
Gameplay & 3D Systems: Player Controls, Weapon Systems, Enemy AI, Camera Systems, Collision Detection, 3D Vector Math, Rotational Interpolation
C++ & Engineering: STL, Object-Oriented Programming, Component-Based Architecture, Debugging, Unreal Automation

TECHNICAL PROJECTS

Protocol Zero | *C++, Unreal Engine 5.8, Enhanced Input, Unreal Automation, Git LFS* | GitHub

- Architected **modular C++ gameplay systems** for weapons, projectiles, health, enemy AI, and wave spawning, exposing gameplay parameters through Unreal Data Assets for rapid gameplay tuning.
- Verified core combat with **47 headless Unreal automation tests** by isolating fire cadence, cursor aiming, damage grace windows and wave scheduling as **engine-free**, **time-injected C++**.
- Implemented **navmesh-driven enemy AI** in C++ that paths around cover and attacks on cooldown, by writing **custom AI controllers** and object collision channels that rule out friendly fire.
- **Debugged a silent C++ AI pathfinding failure** using navmesh projection queries, traced the **root cause** to an editor navigation build lock, and restored reliable pursuit around all 8 cover blocks.
- Caught a projectile damage bug before release by writing **3 world-simulation regression tests** that fire real projectiles at enemies, cover and the player in a headless test world.

Vyxxen | *C++, Unreal Engine 5.8, Enhanced Input, Unreal Automation, Git LFS* | GitHub

- Built a 3D isometric arcade shooter in Unreal Engine 5.8, implementing **custom C++ flight controls**, camera synchronization, weapons, and collision mechanics.
- Implemented deterministic ship-camera synchronization using a shared route clock and **explicit tick ordering**, maintaining consistent camera positioning during movement.
- Implemented a custom **C++ 3D flight model** using **vector mathematics**, **rotational interpolation**, and sweep-based collision detection to support steering, climbing, pitch, and banking.
- Designed **reusable weapon and armor components** that work on any actor, including **frame-rate-independent fire** and an aim-cursor decal projected onto whatever surface the next shot would hit.

Mini Dungeon Adventure | *C++, STL, Visual Studio Code, Git* | GitHub

- Developed a turn-based dungeon crawler in C++ using inheritance and enum-driven item effects.
- Implemented level-scaled enemy encounters using weighted random selection and progression-based monsters.
- Used **STL containers and algorithms** (std::map, std::find_if, std::sort) for inventory management, item lookups, and persistent leaderboard rankings.

EXPERIENCE

UBC Game Development Club | Gameplay Programmer **Sept 2026 — Present**

- Won **1st place** among 10 teams in Battle of Concepts with a boss-rush action game concept selected for development by a 10-person team.
- Collaborated with designers and fellow programmers to translate combat design requirements into reusable GDScript gameplay systems, incorporating playtesting feedback to refine combat interactions.

AllSet Learning | Math & English Tutor **Oct 2025 - Present**

- Earned management recognition as a **top-performing mathematics tutor** by tutoring students and communicating detailed progress and learning needs.